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Integrating gamification in design research: a pedagogical approach for design education in India

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Abstract

Gamification is gaining prominence in design education, offering innovative methods to enhance student engagement and develop critical competencies such as collaboration, creativity, and problem-solving. This study investigates the integration of gamification into a first-year Design Research course at ATLAS SkillTech University, Mumbai, India, over two academic years—Fall 2023 and Fall 2024. The objective was to address the challenges of traditional pedagogies, which often struggle to engage students in research-focused courses, by creating an immersive, student-centric learning experience. A mixed-method approach was employed, combining surveys and personal interviews with 259 respondents, including students and faculty. Specific gamification elements were incorporated into the curriculum to teach the research process, prioritizing active participation and engagement. The revamped course was evaluated to understand the impact of these elements on learning outcomes. Results indicated significant improvements in student engagement, peer collaboration, and creativity, highlighting the potential of gamification to transform research-based learning in design education. Specifically 46% of students found gamified elements such as integration of snakes and ladder engaging. While 30% of the students found the game very engaging indicating that 76% responded positively to the approach. These findings underscore the significance of integrating gamified strategies to meet the diverse needs of modern learners and enhance educational innovation. The study advocates for gamification as a tool to prepare future designers for a rapidly evolving industry while enriching the discourse on adaptive pedagogical practices.

Keywords Gamification, Design education, Student engagement, Pedagogical approaches, Research

1 Introduction

Design education often struggles to engage first-year undergraduate students effectively, particularly in research-focused courses where theoretical concepts can seem abstract and disconnected from practical application. Traditional Pedagogical methods struggle to accommodate diverse learning styles- whether visual, auditory or sensor - leading to disengagement and passive learning [11]. This issue is particularly pronounced in design education, where research is highly individualised and requires innovative teaching approaches to maintain student motivation and participation.

While gamification has been extensively explored in STEM education, its application in design research pedagogy, particularly in India, remains underexplored. By integrating gamified strategies, this study aims to bridge this gap by investigating how game-based learning can enhance student engagement, foster creativity, and promote active participation in research-based design education [14].

The field of design education has transformed significantly over the decades. Traditionally rooted in theoretical knowledge and technical skills, design programs often employed passive learning methods. Designers have different types of challenges and tasks to fulfill; however, students are not usually adequately prepared for these challenges due to traditional pedagogies (Meyer and Norman 2020). The need for educational frameworks that nurture critical thinking, collaboration, and practical skills becomes increasingly evident, as the industry develops. The concept of modern education emphasizes skills such as critical thinking, creativity, teamwork, communication, and socio- cultural awareness, essential competencies for navigating the intricacies of the rapidly changing design industry. Similar to immersive gamification strategies, 3D visualization tools have also demonstrated the potential to improve learning outcomes through enhanced motivation and engagement, as evidenced in Samala et al.'s [15] study using the Hedonic Motivation System Adoption Model in higher education. Contemporary learning theories propose that learning is most impactful when it involves active engagement, hands-on experience, contextual relevance, problem-solving, and timely feedback [19]. Achieving these objectives without technology is feasible, the integration of information and communication technologies (ICTs) has sparked interest in exploring new pedagogical methodologies that can enhance learning experiences Huang et al. [7].

Despite growing recognition of the need for innovative practices, significant gaps persist in design education. Many graduates lack the critical skills necessary for career success. A study by Urh et al. [19] highlights that the transition to university education introduces increased complexity, with students becoming more aware of their field's importance. The contemporary instructional methods may not adequately engage students or promote the application of information in the actual world, which raises questions about their efficacy.

The use of game-design elements in academics to act as a catalyst for student engagement and motivation. This concept has gained recognition in education, as a powerful tool for transforming learning experiences. Key principles of gamification, including competition, rewards, feedback, and achievement, can drive student motivation and foster a sense of regulation in the learning process.

Research shows that gamification significantly impacts student engagement and learning outcomes [3]. Integrating points, badges, and leaderboards can create an environment of achievement and encourage students to strive for excellence [6]. In design education, where creativity and collaboration are paramount, gamification fosters an environment that motivates students to apply theoretical concepts in practical scenarios [2].

The integration of gaming elements into design education, particularly for a course like design research, presents numerous benefits. It encourages hands-on learning by enabling students to interact with the course material as gamified activities can develop essential skills such as problem solving, critical thinking, and collaboration, qualities increasingly sought in the design industry [3].

Moreover, gamification can bridge the gap between theory and practice. Design projects can include gamified challenges that mimic real-world scenarios, prompting students to think creatively and collaborate. Such experiences enhance engagement and prepare students for the complexities of their future careers [17].

However, integrating gamification into design education poses challenges. Educators may lack familiarity with gamification strategies, necessitating targeted professional development for effective implementation [2]. Additionally, there is a risk of superficial engagement if gamification is over relied upon, overshadowing the importance of meaningful content [5].

A balanced approach is essential. Educators must thoughtfully integrate gamified elements to complement substantive learning experiences. This requires careful consideration of how gamification can enhance rather than distract from the learning outcomes of the course.

This study investigates effective gamification integration in design research courses for first year undergraduate students in semester 1 for two consecutive years, focusing on its potential to enhance student involvement and learning outcomes. By examining the experiences of students and faculty, this research identifies best practices for implementing gamification in enriching student engagement and achieving the learning outcomes of the course.

Ultimately, this study aims to transform design education, particularly in context to research as a subject, guiding students to deal with the complexities of the modern design industry while being able to apply research effectively.

2 Research objectives

To guide this study on the integration of gamification into first-year design research education, the following research questions were formulated:

- How does gamification influence student engagement in first-year undergraduate design research courses?
- Which gamified strategies are most effective in fostering creativity and peer collaboration among design students?
- How does gamification help challenge and reshape students' preconceived notions about the nature and purpose of design research?
- In what ways do individual learning styles influence students' engagement with and response to gamified research activities?
- How do gamified activities impact students' perceptions of the relevance and applicability of research in design education?

3 Proposed solution

To overcome the challenges faced in engaging first-year students in design research courses, the proposed solution was to implement a pilot study for the 2023–2024 academic year that incorporates elements of gamification into the existing curriculum. The new pedagogical approach aims to transform conventional teaching methods by making research more interactive and relevant, thereby fostering higher levels of student motivation and participation.

4 Literature review

The literature on gamification highlights its increasing relevance across various sectors, with education being a notable emerging trend. However, while gamification has been widely explored in STEM and business education, its application in design research pedagogy, particularly in the Indian context, remains underexplored. This review synthesizes key studies to identify trends, limitations, and gaps in the literature, positioning our research within the broader academic discourse.

4.1 Systematic mapping of gamification in research education

Dicheva et al. [4] conduct a systematic mapping study of empirical research examining the effects of gamification in educational contexts. They propose a framework categorizing research findings into areas such as gamification design principles, game mechanics, and evaluation. While their framework provides a foundation for effective implementation strategy, they highlight a critical gap; the need for further research on how gamification influences both extrinsic and intrinsic motivation in learners. This gap is particularly relevant to design education where motivation plays a crucial role in fostering creativity and self driven research (Fig. 1).

Kabilan and Chuah (2023) highlights how gamification increases student motivation and classroom engagement through a mixed method study conducted at a Malaysian university. They identified key challenges in implementing gamified learning, particularly the need for tech. support and alignment with academic goals. Similarly, Hasan et al. (2019) argue that gamification enhances motivation by fulfilling psychological needs

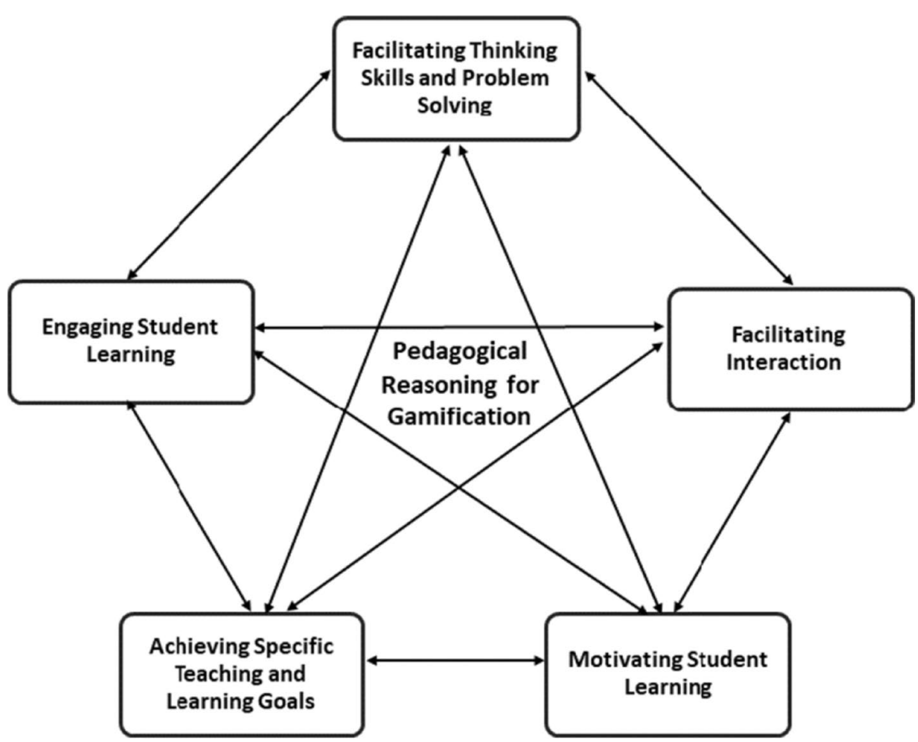


Fig. 1 Multi directional model of pedagogical reasoning for gamification (Kabilan & Chuah 2023)

such as competition and self determination, leading to more spontaneous and voluntary learning. While the study provided valuable insight they primarily focus on structured disciplines. The application of gamification in design research education—where learning is highly individualized and creativity- driven—remains underexplored, requiring further investigation into how specific game mechanics, such as boardgames can foster engagement in this context.

Kofinas [10] investigates the advantages of gamification in teaching research methods within a business management course through a comparison of two cohorts. The gamified curriculum resulted in stronger final grades and higher engagement levels, suggesting its effectiveness to elevate learning outcomes. The pilot study conducted by Mr. Kofinas leads to the development of bespoke software integrating educational pedagogy within a gamified framework. Kofinas's findings provide empirical evidence of improved student engagement and academic performance through gamified methods, informing our research in design education. This contrast raises questions about whether similar improvements in engagement and academic performance can be achieved in design education, where students often navigate research topics in open-ended and highly individual ways.

Despite the notable benefits of gamification in secondary education, research at the postsecondary level remains limited, particularly in fields outside of the STEM field. Many educators implement gamification informally, often lacking the necessary support. Hutson et al. (2022) emphasize the need for sustainable infrastructures and interdisciplinary partnerships to develop well-designed educational experiences. Their qualitative case study on *The Museum of the Lost VR* demonstrates a structured approach to gamification, offering scalable resources for institutions. This highlights the potential for improved learning outcomes through systematic gamification in diverse secondary contexts. This study suggests that while gamification holds promise, its successful application requires well- defined framework, which is currently lacking in design research education (Fig. 2). Meier [12] argues that technology plays a foundational role in supporting 21st-century education by providing a conceptual framework that facilitates collaborative learning, critical thinking, and the integration of digital tools into pedagogical practices.

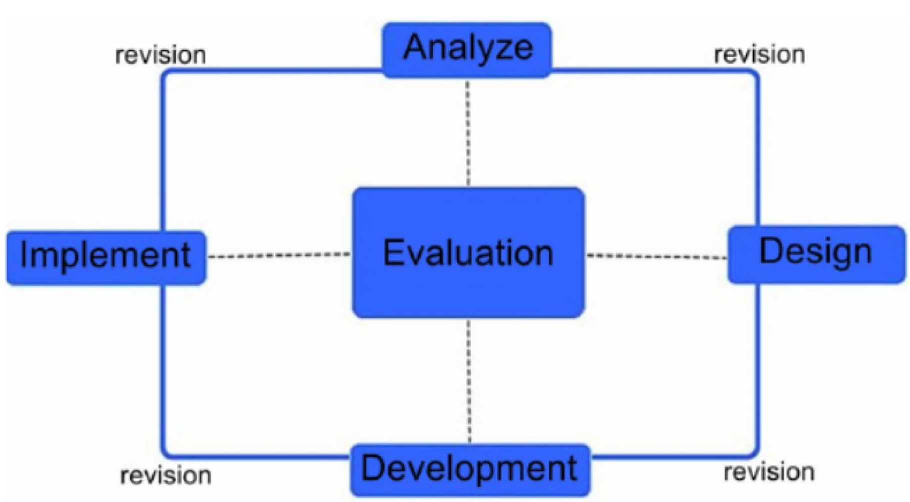


Fig. 2 ADDIE model of instructional design (Hutson et al. 2022)

Zainuddin et al. [21] explored the theoretical frameworks underlying gamification research, identifying the Self determination theory, flow theory and Goal setting Theory as the most frequently used. However, many studies lack explicit theoretical grounding, limiting their ability to generalise findings. Similarly, van Gaalen et al. [20] examined how game elements particularly *assessment* and *conflict /challenge* are embedded in gamified learning environments. Their findings reveal that while these mechanics enhance engagement in health professions education, they are often explicitly tied to pedagogical theories. this gap underscores the need for a theoretical informed approach to gamification in design research education, ensuring that game mechanics align with the students cognitive and creative learning processes. Nasution [13] argues that technology-based teaching and learning in the 21st century have revolutionized education by promoting accessibility, personalized learning, and the development of digital competencies essential for modern learners.

4.2 Academics' perspectives on gamification

Kabilan and Annamalai [9] conducted a mixed-methods study on gamification in higher education, revealing that while many academics incorporate gamification, they lack a cohesive framework for its effective implementation. They identified five key pedagogical themes: motivating students, fostering critical thinking, engaging learners, enhancing communication, and achieving learning objectives. These insights align with our research goal—to develop a structured approach to gamification in design research education, ensuring that game-based learning strategies align with students' diverse cognitive and creative needs. Sokolova, Blaginin, and Shatrova [16] argue that the evolution of STEM education reflects a shift toward interdisciplinary approaches, with current trends emphasizing technology integration, experiential learning, and global collaboration to meet the demands of future-ready education.

Research into gamification is particularly relevant for first-year design students (age group of 17–19), as engaging and interactive pedagogical strategies can enhance their understanding of research methodologies. Implementing gamification in this context can motivate students, foster critical thinking, and help them navigate the complexities of design research. By leveraging the insights from the reviewed literature, our study aims to create a framework that supports first-year students in developing essential research skills through gamified learning experiences. Alsawaier [1] emphasizes that implementing gamification in this context can motivate students, foster critical thinking, and help them navigate the complexities of design research. By leveraging the insights from the reviewed literature, the study aims to create a framework that supports first-year students in developing essential research skills through gamified learning experiences. Tan, Ting, and Tan [18] argue that gamification plays a pivotal role in transforming traditional learning environments by enhancing student engagement, motivation, and participation in higher education contexts.

5 Methodology

The methodology for this pilot study focuses on the integration of gamification in design research for first-year undergraduate bachelor of design (4 years) program at Atlas SkillTech University in Mumbai, India. This research followed a mixed-methodology approach, proposing a gamified pedagogy to teach a research course to design students, keeping in mind their diverse learning styles and backgrounds. Design school students often seek engaging classroom activities that motivate them, and the challenge was to teach research in a more interactive and exciting way while maintaining the same learning outcomes.

The curriculum was reworked to include class activities and games, such as scoreboards and competitions, to encourage participation and interaction. Research theories were integrated through these activities. This revised curriculum was tested on a group of eight students. After initial testing, the curriculum was implemented during the 2023–2024 academic year, followed by minor iterations based on feedback in 2024.

This research paper is based on evidence collected from students and faculty through quantitative and qualitative methods to understand the impact gamification has had in the design research course across two semesters: Fall 2023 and Fall 2024. Additionally, the integration of gamification into the curriculum was informed by secondary data, research, and inputs from the faculty team. This study is grounded in two interrelated learning theories:

Self-Determination Theory (SDT): Emphasizes intrinsic motivation driven by autonomy, competence, and relatedness. Gamified elements were designed to empower students to make choices (autonomy), overcome challenges (competence), and collaborate with peers (relatedness).

Flow Theory: Describes optimal learning experiences as those involving deep immersion and focus. Activities like board games and model-making were structured to induce flow states by balancing challenge and skill.

Figure 3 highlights the curriculum framework for the Fall 2023 and Fall 2024 semesters, integrating gamification pedagogies for every class. A robust research design was employed to ensure the validity, reliability, and potential of findings regarding the effect of gamification on student engagement and learning outcomes. A mixed-methods research approach was utilized to gather comprehensive insights from diverse stakeholders, including students (200) and faculty (10).

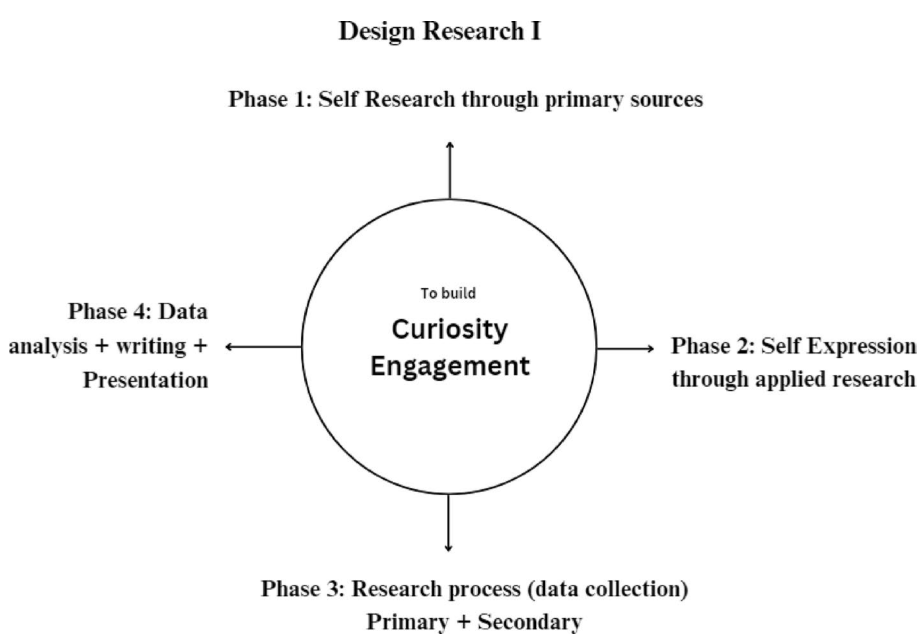


Fig. 3 Design research curriculum framework, ATLAS SkillTech University

5.1 Curriculum and pedagogy

The methodology for the Design Research course was structured around generating curiosity and fostering student engagement, with a four-phase approach that encourages self-driven exploration and applied learning.

Phase 1: Self-Research through Primary Sources: This phase initiates the students' journey into design research by encouraging them to conduct self-research using primary sources. By interacting with real-world data and firsthand experiences, students cultivate curiosity, developing their capacity to observe, question, and understand their environment. The aim is to have students immerse themselves in the subject matter through direct engagement rather than relying solely on secondary literature.

Phase 2: Self-Expression through Applied Research: In this phase, students begin to translate their findings from Phase 1 into self-expressive outcomes. Here, applied research methods come into play, allowing them to utilize their personal insights in the context of design thinking. This phase focuses on creativity, enabling students to connect their research with practice and communicate their findings through visual design outcomes.

Phase 3: Research Process (Data Collection) Phase 3: emphasizes the systematic collection of data, combining both primary and secondary sources. Students deepen their exploration by gathering additional data that supports their initial self-research while learning to critically evaluate secondary sources. This dual-source approach ensures a more robust and clear understanding of their research topic.

Phase 4: Data Analysis, writing, and presentation In the final phase, students are tasked with analyzing the data collected throughout the previous phases. This includes interpreting patterns, drawing conclusions, and integrating their findings into a narrative for further writing an article. The phase culminates in writing and presenting their research, encouraging students to refine their communication skills and articulate their design research process and outcomes clearly and effectively.

Each phase is designed to build upon the last, fostering a research culture grounded in curiosity and self-driven exploration, critical skills for any budding designer. The methodology not only helps students develop a solid foundation in research but also nurtures their ability to express, analyze, and communicate their ideas in both academic and applied settings.

5.2 Ideation

The ideation phase involved brainstorming and conceptualizing potential gamification strategies tailored to design education. Workshops and focus group discussions were held with faculty and students to generate ideas for incorporating gamified elements into the curriculum. This collaborative approach ensured that the proposed strategies were relevant and aligned with the needs and preferences of the target audience.

5.3 Gamification design

5.3.1 Snakes and ladders game

A gamified learning experience was created using the classic "Snakes and Ladders" format, where each class of 30 students were provided with A3 sheets to create a personalized "Snakes and Ladders" game. Instead of a traditional dice mechanism, students illustrated their strengths and weaknesses on the board. They were first asked to write a brief reflection about themselves, identifying key strengths (represented as ladders) and weaknesses (represented as snakes). This exercise encouraged self-awareness and allowed students to visually interpret their personal attributes, enhancing their understanding of how these qualities impact their learning and design process for which were provided one week's time. However the evaluation criteria for snakes and ladders were based on creativity in the board design and self reflection. Additionally engagement levels through class participation and discussion was also considered as a major criteria.

5.3.2 Self-portrait model-making

In this activity, a class of 30 students used various materials, such as clay, wood, and other craft supplies, to create three-dimensional representations of themselves. The models reflected their likes and dislikes, enabling students to express their identities creatively. Participants were encouraged to use photographs as primary evidence, while the models served as secondary artifacts to support their self-exploration. This hands-on experience fostered critical thinking and creativity while also allowing students to engage in research by reflecting on their personal experiences and preferences for which were provided two weeks time. However the evaluation criteria for self Portrait and Model making were based on depth of personal insights. Additionally presentation, creativity and active class participation were also considered.

5.4 Development

Once the gamified activities were conceptualized, they were integrated into the design curriculum. Faculty training sessions were held to equip educators with the skills and understanding needed to effectively implement and evaluate these gamification strategies. This step ensured that the integration was coherent and aligned with the educational objectives of the program as seen in Table 1.

5.5 Data collection

The data collection process consisted of both primary and secondary methods. A literature review was conducted to gather foundational knowledge on gamification, its principles, and its applications in education. Research papers, articles, and case studies were reviewed to identify best practices and previous findings related to gamification in design education. This secondary data collection provided a contextual framework for the study and highlighted existing research gaps.

For primary data collection, an online survey was conducted with 259 students, followed by a focus group discussion and faculty interviews, designed to gather perspectives from participants regarding their awareness, perceptions, and experiences with gamification. The survey included closed-ended questions employing a Likert scale to quantify preferences and attitudes toward various gamification elements.

5.6 Evaluation

The curriculum was piloted in Fall 2023 and revised based on faculty and student feedback before implementation in Fall 2024. Key adaptations included:

1. **Adjusting activity durations** (Snakes and Ladders extended from one to two sessions to allow deeper self-reflection).
2. **Refining assessment rubrics** for clearer evaluation of engagement levels.

Table 1 Snakes and ladders and self-portrait work by students after implementing in the design research curriculum

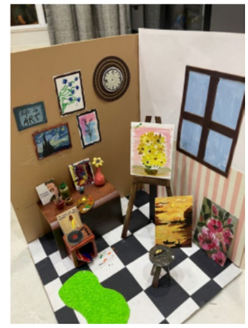
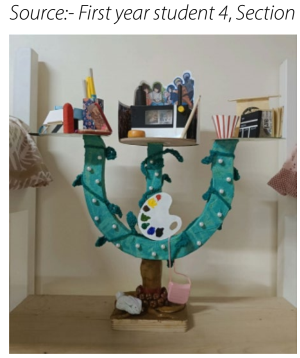
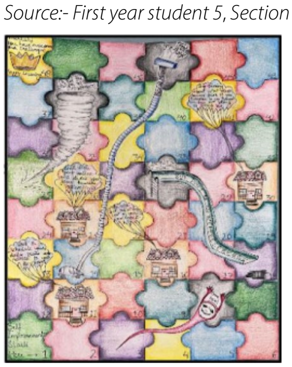
Snakes and ladder	Self portrait
 *Source:- First year student 1, Section Fitbit, ATLAS*	 *Source:- First year student 2, Section Fitbit, ATLAS*
 *Source:- First year student 3, Section Fitbit, ATLAS*	 *Source:- First year student 4, Section Tesla, ATLAS*
 *Source:- First year student 5, Section Tesla, ATLAS*	 *Source:- First year student 6, Section Tesla, ATLAS*
 *Source:- First year student 7, Section Tesla, ATLAS*	 *Source:- First year student 8, Section Tesla, ATLAS*

Table 1 (continued)

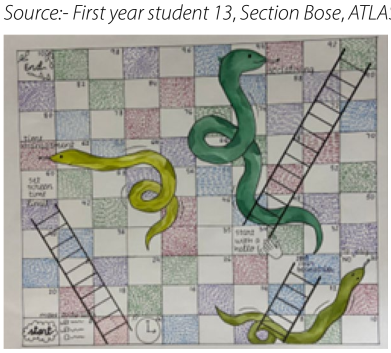
Snakes and ladder	Self portrait
	
Source:- First year student 9, Section Bose, ATLAS	*Source:- First year student 10, Section Bose, ATLAS*
	
Source:- First year student 11, Section Bose, ATLAS	*Source:- First year student 12, Section Bose, ATLAS*
	
Source:- First year student 13, Section Bose, ATLAS	*Source:- First year student 13, Section Bose, ATLAS*
	

Table 1 (continued)

Snakes and ladder	Self portrait
Source:- First year student 14, Section Bose, ATLAS	*Source:- First year student 15, Section Bose, ATLAS*

Table 2 Evaluation criteria for student outputs

Criterion	Indicator
Creativity	Unique visual elements, design coherence
Self-Reflection	Articulation of personal learning, honest self-assessment
Collaboration	Active group participation, peer feedback
Research Linkage	Evidence of connecting play-based insights to research work

Assessment Criteria Evaluation criteria for student outputs included:

- **Creativity**: Originality, expression, coherence.
- **Self-Reflection**: Depth of personal insight, connection to research goals.
- **Participation**: Engagement in discussions and teamwork.
- **Application**: Ability to link activities to design research processes.

3. **Introducing peer reviews** to enhance collaborative learning (Table 2).

The evaluation process involved assessing the effectiveness of the gamification strategies in enhancing student engagement and learning outcomes. The online survey, initially distributed, was utilized again to gather feedback on the implemented activities. The questionnaire was structured into four main sections:

1. Demographic Data: Basic information about respondents, including their roles (students or faculty) and their experience in design.
2. Engagement with Gamification: Questions focused on participants' experiences with the "Snakes and Ladders" activity and self-portrait model-making, assessing how these activities influenced their engagement and understanding of design concepts.
3. Perceived Benefits: Participants rated the perceived effectiveness of the gamified activities in enhancing their creativity, self-awareness, and collaborative skills.
4. Impact Assessment: This section gathered qualitative feedback on observed changes in motivation and engagement levels resulting from the gamified learning experiences.

5.7 Survey and analysis

To ensure comprehensive representation of perspectives, a stratified random sampling technique was employed to gather responses from various groups within the design education community. A total of 259 respondents participated in the survey, allowing for robust data analysis. After collecting the data, it was compiled, tabulated, and analyzed using statistical methods to identify trends and correlations related to the effectiveness of the gamification strategies.

5.8 Tools for analysis

The data analysis process utilized both qualitative and quantitative methods. Statistical software was used to analyze survey results, providing insights into participant preferences and experiences. Qualitative feedback from open-ended interview questions was thematically analyzed to capture nuanced perspectives on the impact of gamification on learning.

Based on the research findings, gamified learning modules were developed that incorporated the "Snakes and Ladders" activity and self-portrait model-making into the design research curriculum. The focus was on creating engaging and interactive content that resonated with students while aligning with contemporary design practices. The biggest challenges addressed during this phase included ensuring that the gamified elements were not only engaging but also pedagogically sound and relevant to the curriculum.

6 Results and discussion

The survey results on the integration of gamification into the design research course for first-year undergraduate students reveal several positive outcomes that underscore its effectiveness as a pedagogical approach.

Primary research

6.1 Qualitative

6.1.1 Student perspectives

In focus group discussions, students expressed that gamification significantly altered their perceptions of the Design Research course. Many had previously found research to be tedious and overly theoretical. The introduction of games, such as Snakes and Ladders, allowed them to engage with concepts in a more interactive and creative manner.

These activities introduced spontaneity into their classes, making the learning process more enjoyable. Students felt more inclined to express their thoughts, enhancing their overall learning experience.

The positive feedback from students encourages facilitators to explore new games that can effectively support course outcomes. There is also interest in incorporating self-gamification strategies, enabling students to create their own games. This approach can deepen their understanding of the subject while developing critical thinking, creativity, and collaboration skills (Fig. 4).

To further investigate the relationship between student engagement and learning outcomes, a Chi-square test of independence was conducted. The test examined whether there was a statistically significant association between students' perceived engagement with gamified activities (such as Snakes and Ladders) and their understanding of key research concepts. The results revealed a strong and statistically significant association, $\chi^2(1, N=259)=46.20$, $p<.001$. Students who found the gamified activities engaging were significantly more likely to report that these strategies helped them understand research concepts better. This supports the study's assertion that gamification positively impacts conceptual learning by enhancing motivation and cognitive connection as seen in Table 3.

In summary, gamification offers a hands-on approach to learning, fostering engagement and collaboration between students and faculty. By making the educational experience more interactive, gaming reinforces key concepts and improves retention.

6.1.2 Faculty perspectives

Gamification activates student engagement by transforming the learning experience. It has proven effective in enhancing information retention and recall. While it can be applied across various disciplines, it's crucial to consider the specific context and long-term benefits of gamification rather than focusing solely on immediate outcomes.

Please share if you have any personal experiences that reflect on how the gamification teaching methods helped you in learning design research.

64 responses

It made me think about I can express myself and my thoughts in visual form. That is communicate with others without using words

It was all great

It just kept me motivated and I couldn't stop working if I was in the flow it was really engaging. I always love design research classes!

The activities were helpful in identifying and applying key research elements like primary research, secondary research, impact of opinions and biases as well as the need for objectivity.

The gamification method and classroom activities were fun as we learned through experimenting and implementing the ideas. However, sometime the brief was a bit difficult to understand, but eventually with discussions with the faculty it got better.

the faculty helped us understand what was to be done better. gamification of the assignments made it more fun to learn

Fig. 4 Specific statements from students from the online survey conducted

Table 3 Evaluation criteria for student outputs

Engagement level	Found gamification helpful	Did not find it helpful	Total
Engaged	142	55	197
Not engaged	14	48	62
Total	156	103	259

Additionally, spontaneous gamification, where faculty introduce informal activities outside the structured curriculum, adds a layer of dynamism to the classroom. The impact of gamification on student cognition is notable; for instance, role-playing exercises can encourage participation from quieter students and promote the sharing of diverse cultural perspectives.

Games also play a significant role in fostering connections among students. They create a supportive environment where individuals feel comfortable sharing their ideas and experiences. Strategic use of games, such as chor police or card games, highlights the potential for various approaches to lead to effective learning outcomes. Each student cohort brings unique dynamics, necessitating a tailored approach to maximize engagement.

Indian games, both traditional and contemporary, carry deep cultural significance and historical wisdom. They often reflect the values, philosophies, and social structures of their time. By integrating these rich narratives and teachings into research and design, we can create more meaningful and contextually relevant experiences. These experiences with games highlight community bonds and cooperation, showcasing lessons in teamwork and trust.

Through Gamification we can draw allegorical connections between games and design where we can identify with different perspectives of the navigation, the user, the teammate, the competition etc. This aids in getting a wholesome picture of the situation and where as design researchers we can step in.

Since the introduction of gamification, it has been encouraging to observe students actively exploring new ideas and stepping outside their comfort zones. They are not only absorbing information but also reflecting on their experiences and considering how to apply what they have learned.

So when we work these kinds of concepts into a game, it is when the learning becomes contextual. Curriculum designers and faculties are also learning every day and they are adding to their own experiences.

Integrating concepts into gameplay not only makes learning more engaging but also contextualizes it, allowing players to see real-world applications. As curriculum designers and educators immerse themselves in this process, they continuously evolve their understanding and methods, enriching the educational experience for everyone involved. This dynamic exchange of ideas fosters creativity and innovation in teaching, making the learning journey more relevant and.

However, challenges remain. The challenges to inspiring conversations and governing those conversations aimed at what the thought outcome is. Its challenging when one is working with such a large group of students. At times, the emphasis on fun can distract from the learning objectives, leading to a loss of focus.

6.2 Quantitative

259 total first-year undergraduate design students responded to this survey that assessed their experiences when the gamification pedagogical approach was applied to the design research course. The questionnaire was designed to evaluate the impact of gamification on their motivation and engagement levels as seen in Fig. 5.

46% of students agreed that elements of gamification, such as snakes and ladders, were somewhat engaging, whereas 30% agreed that these were very engaging. This suggests that more than half of the students found these elements engaging. Snakes and ladders was applied as an outcome of one of the activities that addresses individual researcher's learning strengths and weaknesses so each can identify and find their unique way of conducting research in design. The students found that the use of snakes and ladders found it engaging, and it significantly improved their learning experience. These elements might have encouraged the students to participate and be more involved and focused. This also indicates that a blend of learning with play has helped students in interacting and engaging more. These students (76%) have responded positively to the application of gamification in the design research course, which also made the course more engaging.

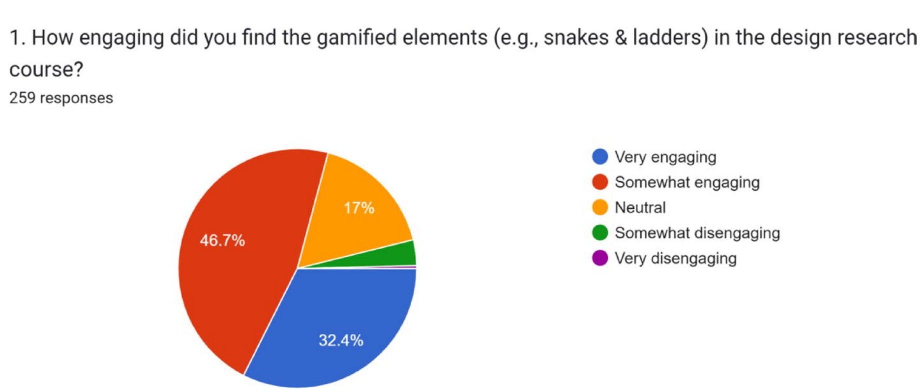


Fig. 5 Screenshot from the survey conducted with first year students in Fall 2024 semester

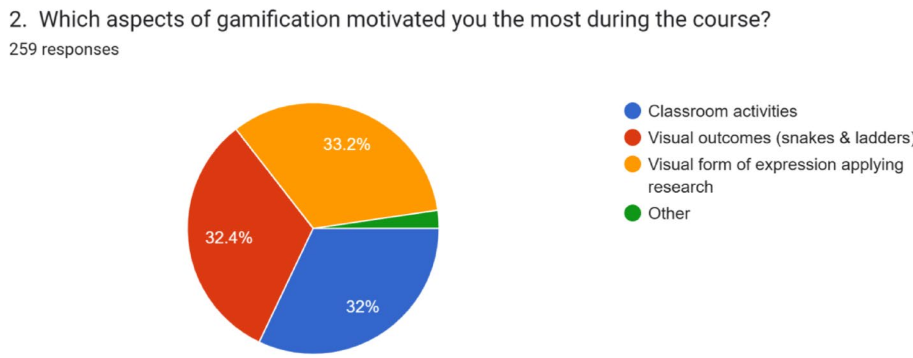


Fig. 6 Screenshot from the survey conducted with first year students in Fall 2024 semester

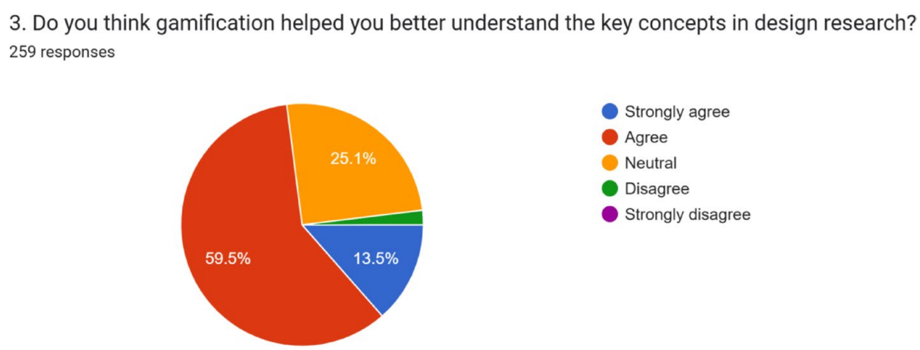


Fig. 7 Screenshot from the survey conducted with first year students in Fall 2024 semester

As seen in Fig. 6, classroom activities and visual expression of the research process motivated the students more than any other gamification strategy. 33.2% of students agreed that both aspects encouraged and motivated them during the course. 30.5% of students also agreed that the visual outcome (snakes and ladders) that was integrated within the course as an outcome of the research process was motivating them during the course. This also suggests that largely students did feel motivated during the course when different gamification strategies were applied. These results clearly indicate that a combination of visually engaging outcomes, tools, and classroom activities has enhanced the motivation level of students.

Figure 7 clearly states that 60.4% of students agree that gamification helped them understand the key concepts in design research better. The majority of students think that the pedagogical approach integrating gamification has helped them learn about the fundamental concepts in research better.

The survey as per Fig. 7 strongly suggests that the majority of students do support the integration of gamification into design education. An important point to note is also that this strategy is well received by the first-year undergraduate students, that is, between the age groups of 17 and 19, which also suggests that gamification as a method is well supported by these students because it helps them in not just learning theoretically but also in understanding the practical application of these theories in design. Understanding complex processes in research in the first year will definitely enhance their learning experience, which has an impact in the following specialization years as well.

While most students responded positively to the integration of gamification, faculty observations indicated that maintaining focus amidst engaging activities was sometimes a challenge. Spontaneous gamification elements also added dynamism to classroom interactions, reinforcing learning objectives through participation.

7 Limitations and future directions

While this study demonstrates the positive impact of gamification in design education, the findings are limited to a single institution. Future research could explore longitudinal studies or cross-institutional comparisons to validate these findings. Additionally, future implementations should consider structured frameworks to ensure gamification aligns seamlessly with pedagogical objectives.

While the study offers significant insights, several limitations merit further attention. The research was conducted at a single institution, which may affect the broader applicability of the findings. To enhance generalizability, future studies should include multi-institutional or cross-regional samples, ideally covering diverse cultural and socioeconomic backgrounds within India. The integration of Indian traditional games, while culturally resonant, was not deeply analyzed for its socio-cultural learning implications. Exploring how culturally specific game elements influence student motivation and identity formation would add depth to future research. Moreover, while chi-square analysis provided initial evidence of statistical significance, further application of multivariate analyses such as ANOVA or regression could offer a more robust understanding of how different gamified activities correlate with diverse learning outcomes. A longitudinal study design would also help determine whether the observed positive effects of gamification are sustained over time, especially across different academic years or course levels.

8 Conclusion

The transforming landscape of design education in recent years has created a necessity to design curriculum that not only engages students but also helps them in learning better. Gamification as a pedagogical approach towards a course like design research for undergraduate first-year students helped in increasing student engagement. Elements of gamification such as rewards, competition, scoreboards, and familiar game structures (snakes and ladders) as outcomes were integrated within the curriculum, and it definitely helped students understand the fundamental concepts of research better. It also helps the educators in building a more engaging and motivating learning environment.Hung, Yang, and Hung [8] highlight that the motivation to gamify the learning process stems from the alignment of game design with self-determination theory and flow theory, emphasizing the importance of autonomy, engagement, and a sense of achievement in sustaining learner interest.

In conclusion, this study also emphasizes the importance of approaching research from a new lens, particularly from a pedagogical perspective. The survey results indicate that gamification helped the majority of students understand the key concepts of research. Research in design education has to be taught in ways that it addresses the disconnect between concepts and practical application in design. Different gamified activities and outcomes helped students to actively engage with course content and also develop a deeper understanding of the research processes. One of the most important aspects was acknowledging individual learning styles and accommodating their diverse needs in the design curriculum.

There were many challenges while implementing gamification as a pedagogy, such as educator's training and a balanced approach to enhance student learning. The first pilot run was in the academic year 2023-24, followed by 2024-25, which also incorporated the feedback and learnings from the first year. This study presents the positive impact and growth in student engagement in two consecutive academic years after applying gamification as a pedagogy. The study's generalizability may be limited by its single-institution focus.

This research holds relevance looking at the future of design education, particularly in context to research as a course, as it highlights the need for innovative pedagogical approaches to increase student engagement. These results align with emerging calls for experiential learning in creative disciplines worldwide, offering a scalable model for integrating gamification into research pedagogy. Gamification can help the educators create a more inclusive and relevant learning environment, preparing students for design careers where they are able to apply their learning and research theories into the field of design.

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Author contributions
Ishi Srivastava, as the course leader for the Design Research course, conceptualized the study and led the implementation of gamification strategies into the curriculum. She collaborated with Shirali Tyabji in conducting primary research, which included organizing and facilitating a survey, focus groups, and interviews with students and faculty.Heena Sachdeva conducted an extensive literature review and secondary research, providing a strong theoretical foundation for the study. She identified key works and frameworks on gamification and its application in education, ensuring that the research was well-grounded in existing scholarly discourse. In addition, she contributed to the synthesis of findings, aligning them with broader pedagogical practices in design education.Shirali Tyabji took the lead in analyzing the primary data collected, ensuring that the findings were accurately interpreted and effectively linked to the study's objectives. She also collaborated with Ishi during the data collection phase, ensuring a robust and comprehensive research process.All authors collaboratively proofread and edited the manuscript, contributing to its coherence, clarity, and final presentation. This study reflects the shared efforts and expertise of the team.

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Data availability
Data is provided within the manuscript.

Declarations

Ethical statements
This study involved non-invasive research methods such as group discussions, questionnaires, surveys, and interviews, which did not involve sensitive personal data, medical interventions, or pose any potential risk to participants. As per the guidelines of ATLAS SkillTech University, the Institutional Review Board (IRB)/Ethics Committee has determined that formal ethical approval is not required for such studies. The research was conducted in compliance with institutional ethical guidelines, and informed consent was obtained from all participants before data collection.

Informed consent
Informed consent for participation and publication has been obtained from all participants involved in this study.

Consent to publish
We have anonymized all identifying details while retaining the pictures. For participants under the age of 18, consent was obtained from a parent or legal guardian. Additionally, any personal information or images included in the study have been published with the necessary consent to ensure participant privacy.

Clinical trial number
A Clinical Trial Number is not applicable to this manuscript.

Competing interests
The authors declare no competing interests.

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